

# Bottle LoRA off the GSE Multiobject Model — Real-Episode Eval

*Does adding a bottle LoRA to the GSE model help? — base GSE vs GSE + LoRA*

| WITHIN-TRAIN (15 eps) | avg MAE | best ep | worst ep |
|-----------------------|---------|---------|----------|
| base GSE              | 1.09°   | 0.90°   | 1.33°    |
| GSE + LoRA            | 0.53°   | 0.43°   | 0.62°    |

**HELD-OUT generalization (unseen episodes, same camera):**

| held-out [100:299]      | base GSE* | GSE + LoRA |
|-------------------------|-----------|------------|
| horizon-end MAE (deg) ↓ | 1.31      | 8.62       |
| first-step MAE (deg) ↓  | 1.24      | 7.48       |
| within 5° (%) ↑         | 100       | 34         |
| displacement r ↑        | 0.934     | 0.236      |
| direction cosine ↑      | 0.833     | 0.357      |

base GSE = pi05\_farm\_multiobject\_gse (the GSE multiobject model, step-5999). GSE + LoRA = pi05\_farm\_bottle\_lora\_gse (a bottle LoRA trained on multiobject[0:100] on top of that GSE model, via GSEMergeWeightLoader). Eval = open-loop teacher-forced single-shot prediction. WITHIN-TRAINING = 15 of the 100 bottle episodes (30 frames each). HELD-OUT = unseen episodes multiobject[100:299], same camera + task (the farm\_uf850\_bottle set is a different camera config → excluded). \*base GSE trained on ALL these episodes, so its rows are a FIT ceiling, not a held-out number.

## Findings

# base GSE vs GSE + LoRA – does adding a bottle LoRA help?

We compare the **GSE multiobject model** (**base GSE**, `\pi05_farm_multiobject_gse`, step-5999`) against the same model with a **bottle LoRA** trained on top of it (**GSE + LoRA**, `\pi05_farm_bottle_lora_gse`, a LoRA on \multiobject[0:100]` initialized off base GSE via \GSEMergeWeightLoader``). Metric = open-loop, teacher-forced single-shot prediction (the model sees the real observation, outputs a 10-step absolute-joint chunk; we score `\pred[k]` vs the real recorded \state[t+1+k]``; horizon end = 333 ms ahead). This is **prediction fidelity**, not closed-loop task success.

## TL;DR

- **On the 100 bottle episodes the LoRA specialized on, GSE + LoRA is the clear winner.** Across 15 of those episodes it reproduces the demos near-perfectly and **uniformly** – **0.53°** average (range 0.43–0.62°, **no outliers**) vs base GSE's **1.09°**. The LoRA tightens bottle fit ~2x.
- **But the LoRA bought that by over-specializing.** On **other** bottle episodes (`\multiobject[100:299]`, same camera`) GSE + LoRA degrades to **8.62°**, while base GSE still handles them at **1.31°** – those episodes were in base GSE's training, so this is a fit ceiling, but the point stands: base GSE knows the whole bottle distribution; the LoRA traded that breadth for a tighter fit on its 100.
- **Net:** add the LoRA if you only care about those 100 episodes' exact motions; keep base GSE if you need bottle competence beyond them.

## 1. Within-training fit – 15 episodes (`\figs/per_episode_fit.png`, \figs/traj_fit15_gse.png``)  
15 episodes drawn from the 100 the LoRA trained on (30 frames each), scored for both models.

| model             | avg per-episode MAE | best ep | worst ep | outliers ( $>\mu+2\sigma=0.65^\circ$ ) |
|-------------------|---------------------|---------|----------|----------------------------------------|
| base GSE          | 1.09°               | 0.90°   | 1.33°    | –                                      |
| <b>GSE + LoRA</b> | <b>0.53°</b>        | 0.43°   | 0.62°    | <b>none</b>                            |

- **GSE + LoRA reproduces every one of these episodes near-perfectly and consistently** (0.43–0.62°, displacement  $r$  0.984, direction cosine 0.954). The fit is uniform – **no episode is a poor-fit outlier**; it memorized all of them evenly.
- **base GSE is looser (~1.1°)** on the very same episodes – it splits capacity across four objects, so the bottle-specialized LoRA fits tighter. Both have zero outliers; GSE + LoRA is simply ~2x closer.
- The trajectory overlay (`\traj_fit15_gse.png``) shows the LoRA's commanded next-angle sitting on top of the real demo across all 6 joints + gripper.

## Findings (cont.)

## 2. The cost: generalization beyond the 100 (`figs/headtohead.png`, `figs/fit\_vs\_heldout.png`, `figs/ckpt\_sweep.png`)

Probe = unseen bottle episodes `multiobject[100:299]` – same camera + same →box task. (The separate `farm\_uf850\_bottle` set is a different camera config → excluded as a domain shift, not a generalization signal.)

|                         |           |             |                  |                 |  |
|-------------------------|-----------|-------------|------------------|-----------------|--|
| held-out [100:299]      | MAE ↓     | within-5° ↑ | displacement r ↑ | direction cos ↑ |  |
| --- --- --- --- ---     |           |             |                  |                 |  |
| base GSE* (fit ceiling) | **1.31°** | 100%        | 0.934            | 0.833           |  |
| **GSE + LoRA**          | 8.62°     | 34%         | 0.236            | 0.357           |  |

- \*\*The LoRA over-specialized.\*\* GSE + LoRA collapses from its near-perfect fit (0.53°, disp r 0.98) to

8.62° / disp r 0.24 on these episodes – it memorized its 100 rather than learning a broad bottle policy. Direction cosine 0.36 is above chance (0) but far below the 0.95 fit: partial generalization.

- \*\*base GSE handles the same episodes at 1.31°\*\* because it trained on all 299 bottle episodes.

\*This is

a fit ceiling, not a fair held-out comparison\* – but it shows base GSE retains competence across the

whole bottle distribution that the LoRA gave up.

- \*\*Checkpoint selection\*\* (`ckpt\_sweep.png`): if you do use the LoRA, \*\*step-2000 generalizes best\*\*

(8.36° / 38%), not the default step-9999 – "later ≠ better" (held-out MAE peaks at step-6000), the over-fit signature.

## 3. Caveats

- \*\*Open-loop, teacher-forced single-shot prediction – NOT closed-loop task success.\*\*

- \*\*Absolute MAE is flattered by the bounded joint workspace;\*\* displacement r / direction cosine are the load-bearing signals.

- \*\*base GSE's held-out rows are FIT ceilings\*\* (it trained on these episodes), not a clean held-out comparison – included to show retained breadth, not to claim base GSE "generalizes" better per se.

- \*\*No seeds / no error bars / single probe.\*\*

## Bottom line

\*\*Adding a bottle LoRA to the GSE model makes it a near-perfect, uniform reproducer of the 100 episodes

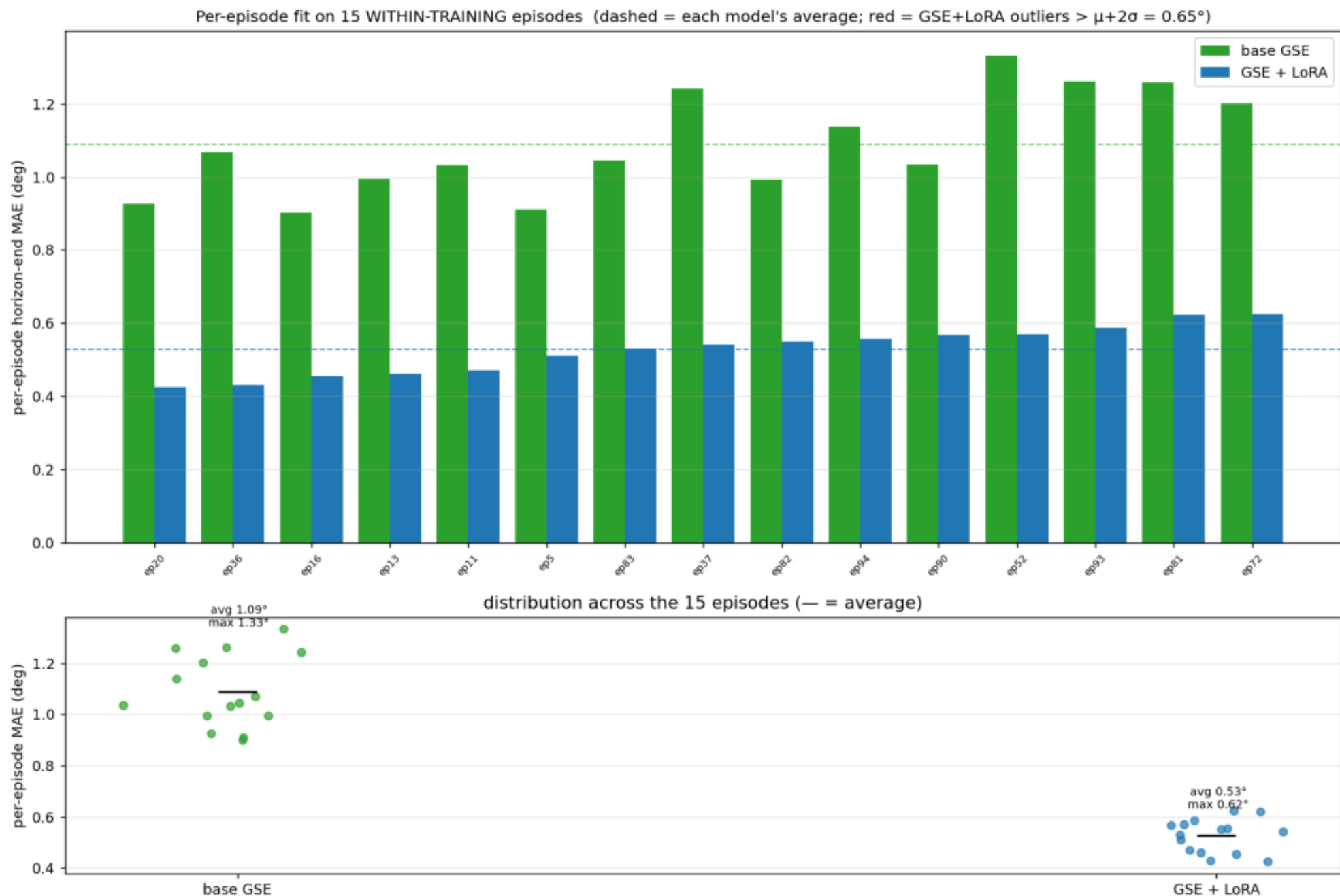
it specialized on (~0.5° vs base GSE's ~1.1°), but it loses competence on bottle episodes outside that

set (8.62° vs base GSE's 1.31° fit).\*\* Use GSE + LoRA when you need the tightest fit on those specific

demonstrated motions; keep base GSE when you need to cover the broader bottle distribution. If you ship

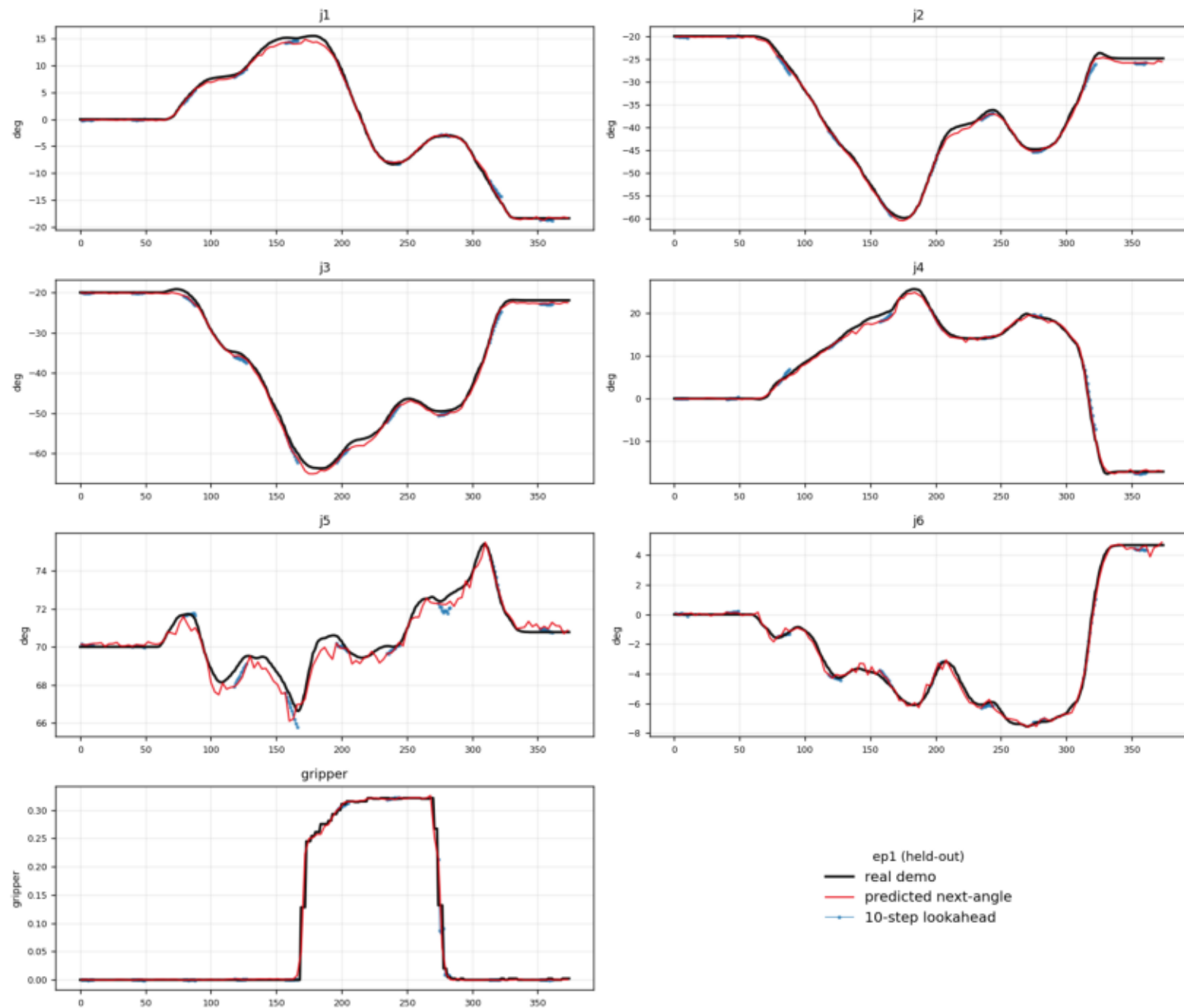
the LoRA, ship \*\*step-2000\*\*.

# Within-training fit — reproducing 15 episodes the LoRA trained on (base GSE vs GSE + LoRA)



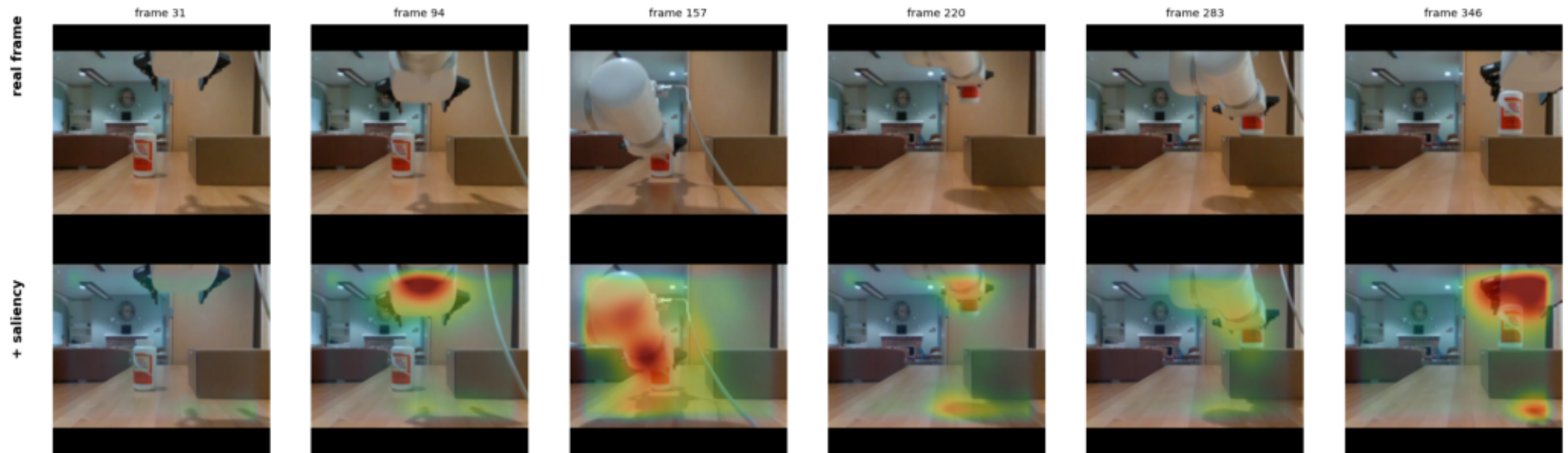
Per-episode fit on 15 episodes the LoRA trained on: per-episode MAE, each model's average (dashed), and GSE+LoRA outliers (red). Lower = closer reproduction of the demo.

predicted next-angle vs real movement on a HELD-OUT bottle episode



GSE + LoRA reproducing a bottle episode it TRAINED on: predicted next-angle (red) vs the real demo (black) — near-perfect tracking on training data.

Where the GSE + LoRA policy looks — episode 1: Picking up the bottle and placing it on the box  
(turbo = pixels whose change most affects the predicted action; SmoothGrad)



What the GSE + LoRA policy looks at: SmoothGrad saliency (gradient of the predicted action w.r.t. the input image) over real frames from a within-training episode — bright = pixels that most change the action. Full episode video: saliency\_episode\_<id>.mp4 in this folder.

What the VLA finds significant — full task string vs the single word "bottle" (episode 1)  
saliency = gradient of the predicted action w.r.t. the image

real frame

frame 46

frame 119

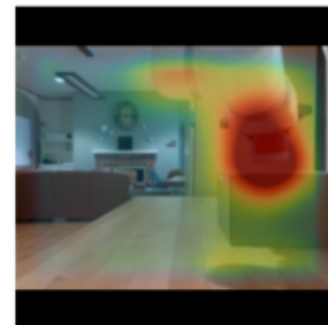
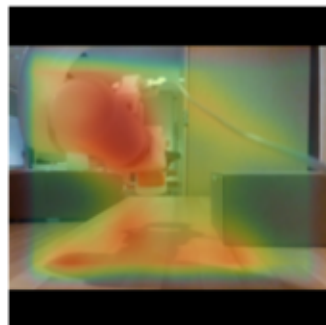
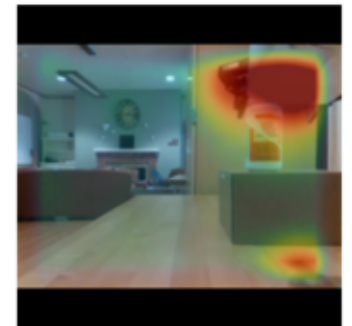
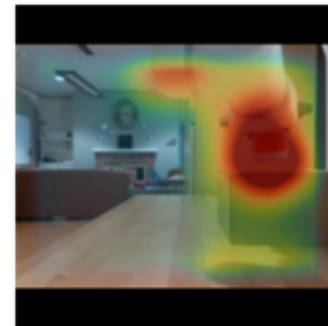
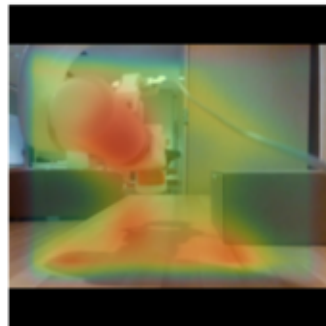
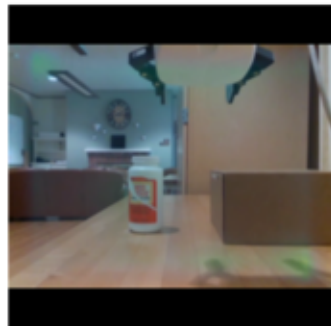
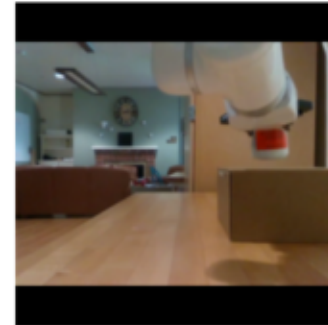
frame 192

frame 265

frame 338

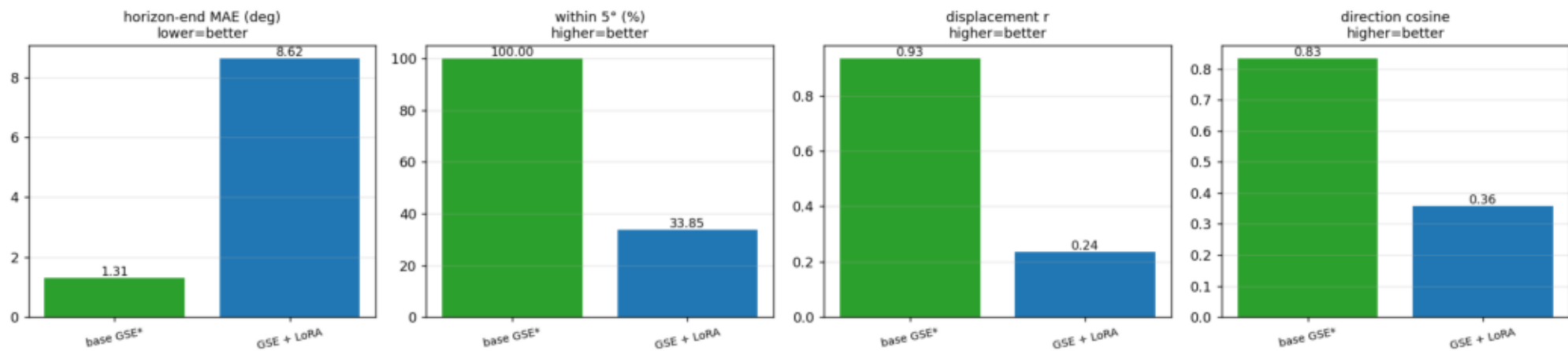
full-task prompt

prompt: "bottle"



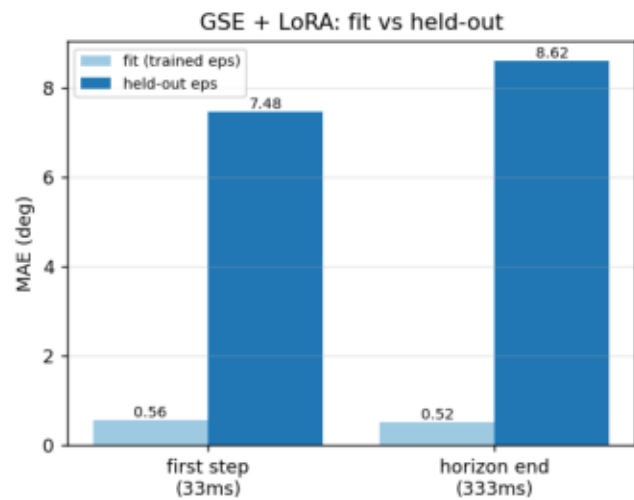
Prompt ablation: what the VLA finds significant when given the FULL task string vs. just the single word "bottle" — same frames, same image, only the language prompt differs. Videos: saliency\_episode\_<id>.mp4 and saliency\_episode\_<id>\_bottle.mp4.

Held-out bottle (multiobject[100:299], same camera) — base GSE vs GSE + LoRA (\*base GSE trained on these episodes = fit ceiling, not held-out)

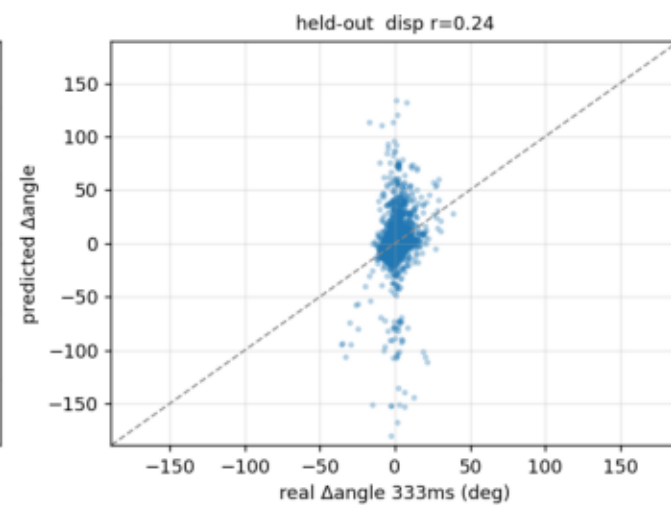
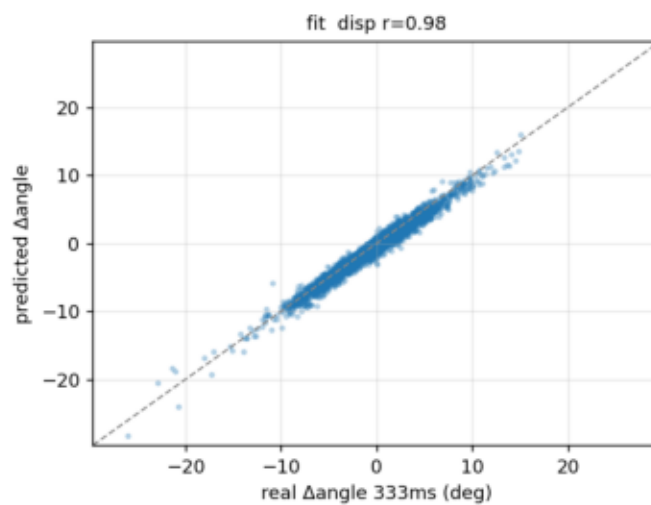


Held-out generalization (multiobject[100:299], unseen episodes): base GSE vs GSE + LoRA (base GSE = fit ceiling).



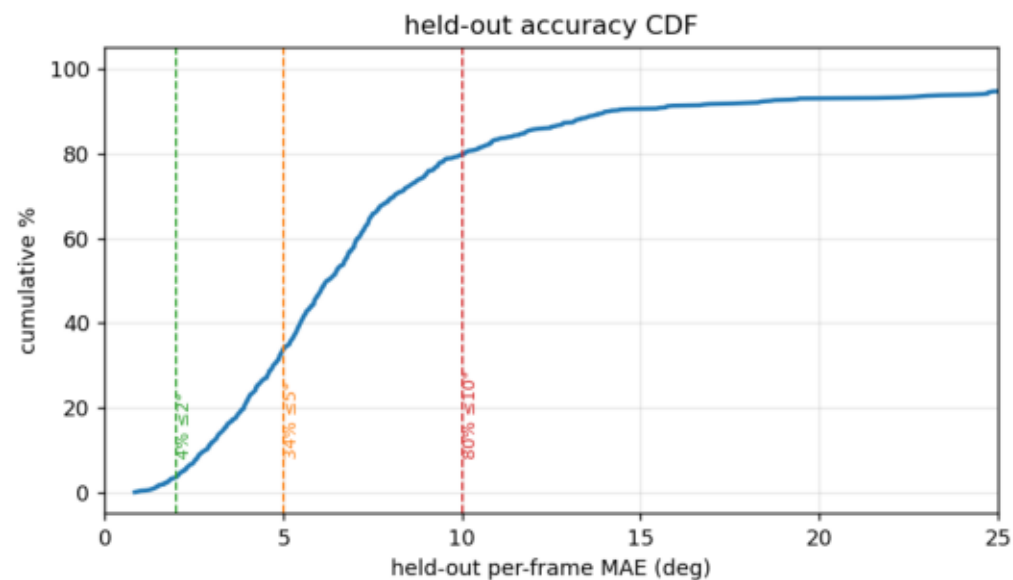
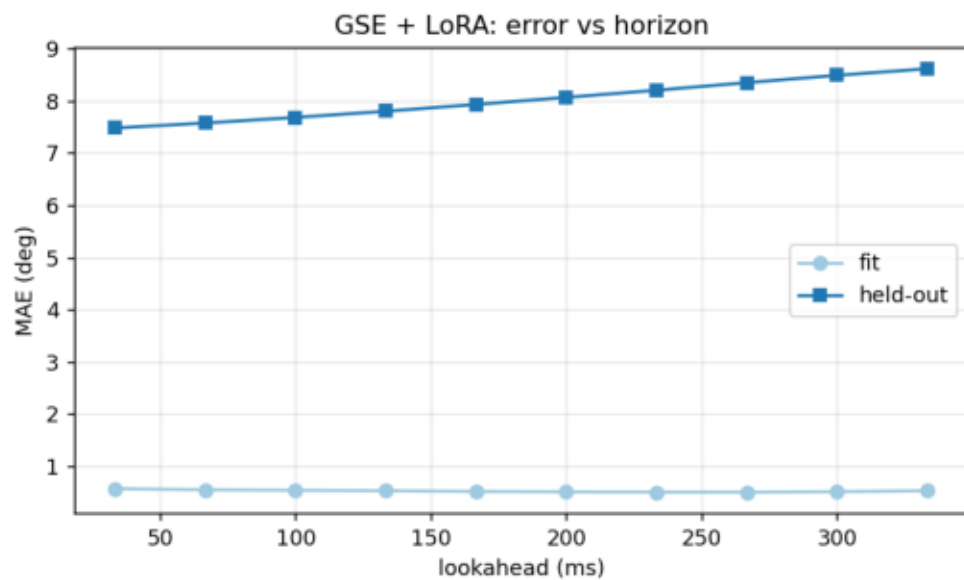


GSE + LoRA — does it track motion on UNSEEN bottle episodes?



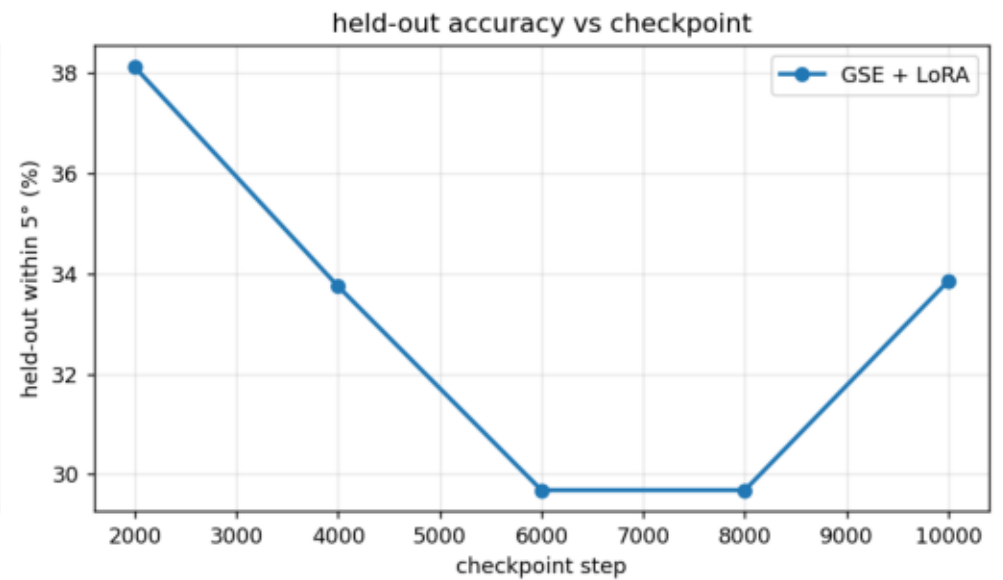
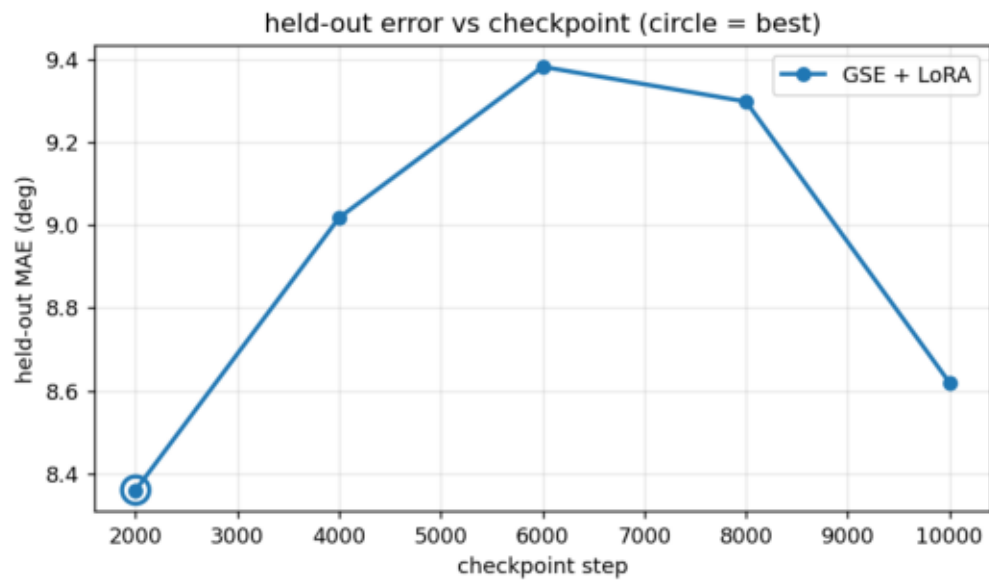
GSE + LoRA: near-perfect on trained episodes, degrades on unseen ones — displacement scatter shows how much motion-tracking survives.

### GSE + LoRA on held-out bottle



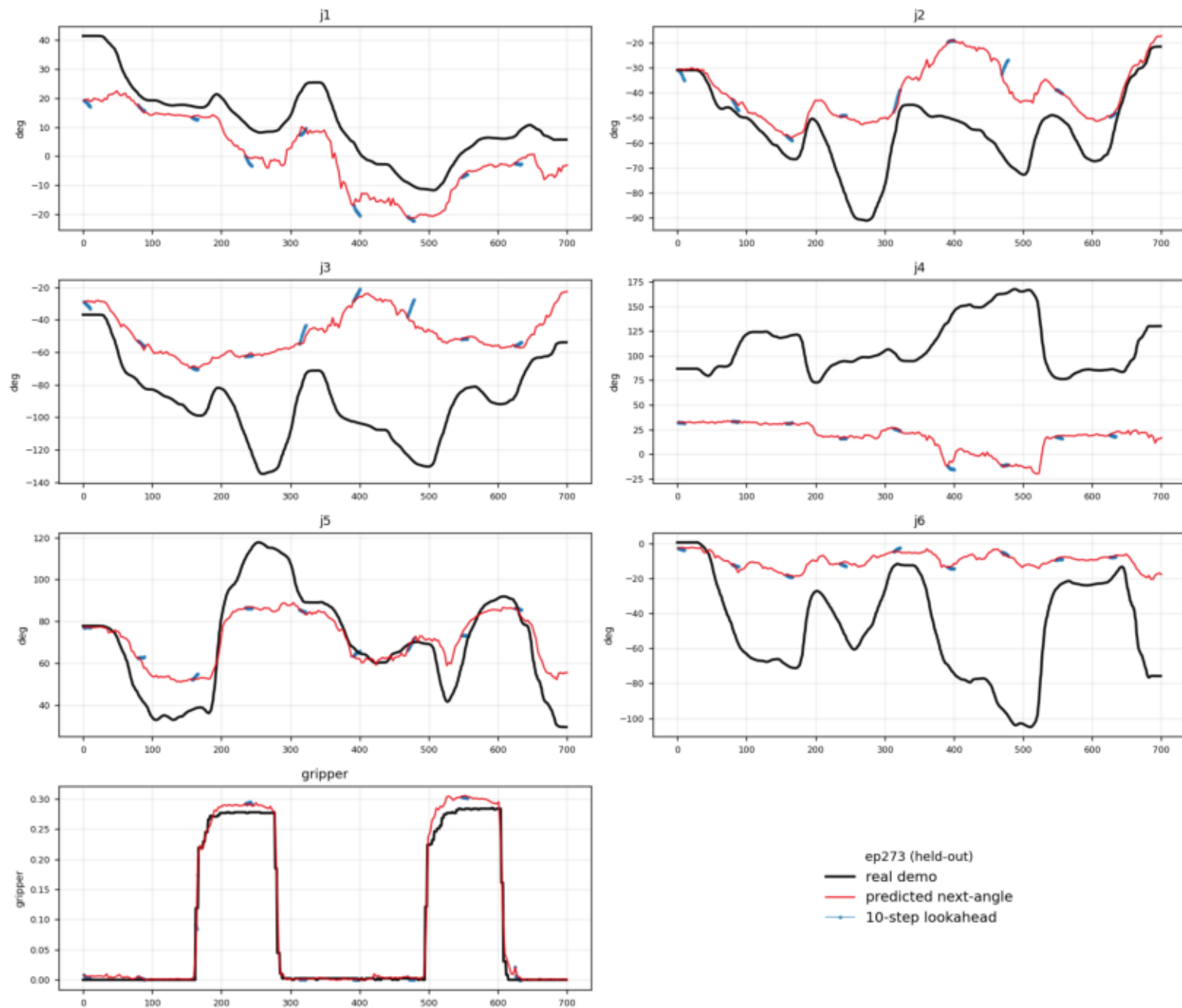
GSE + LoRA: error growth across the chunk + held-out accuracy CDF.

Checkpoint selection on held-out bottle — later  $\neq$  better (the over-fit signature)



Checkpoint selection: held-out error/accuracy vs training step across training steps (the GSE + LoRA).

predicted next-angle vs real movement on a HELD-OUT bottle episode



GSE + LoRA: commanded next-angle vs real movement on a held-out bottle episode.